

MusAI Seminar 5

ML, Public Value, and Institutional and Design Alternatives: the Case of Music Recommender Systems

Georgina Born

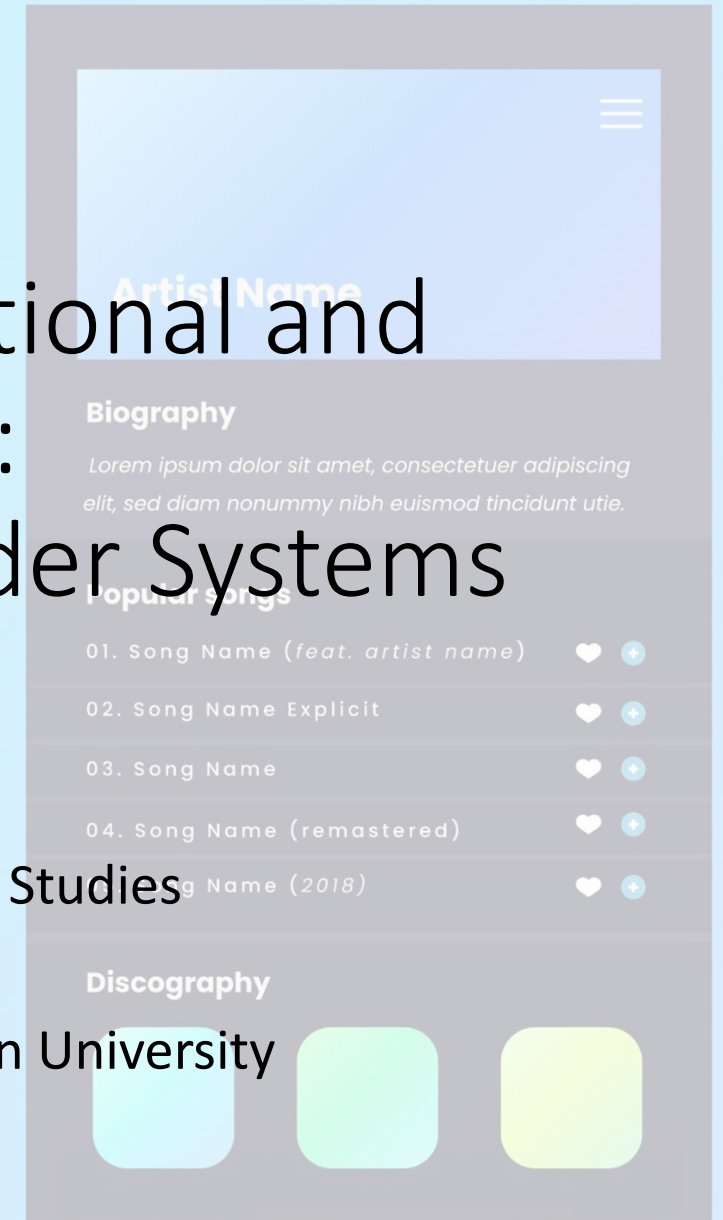
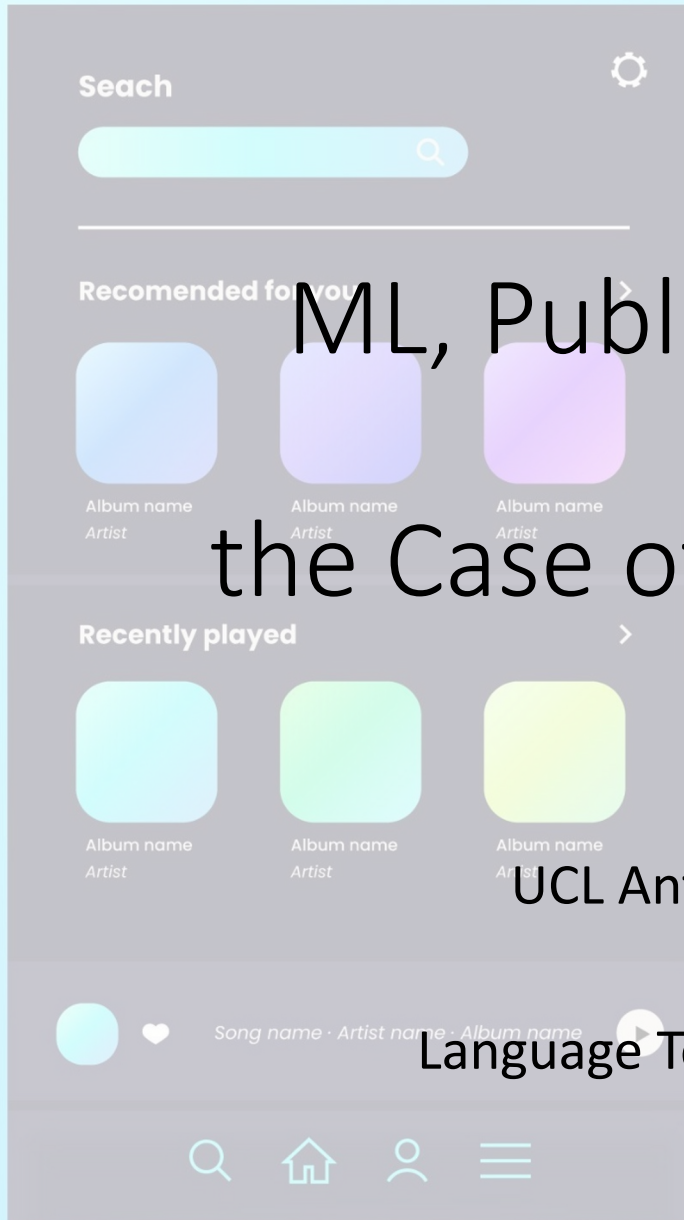
UCL Anthropology and Institute of Advanced Studies

Fernando Diaz

Language Technologies Institute, Carnegie Mellon University

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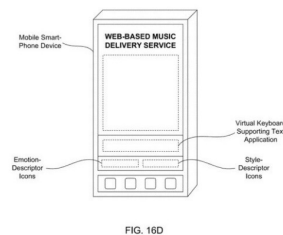




WP1a: Constructing Saliency: Who is the Subject of Musical Machine Listening Research?



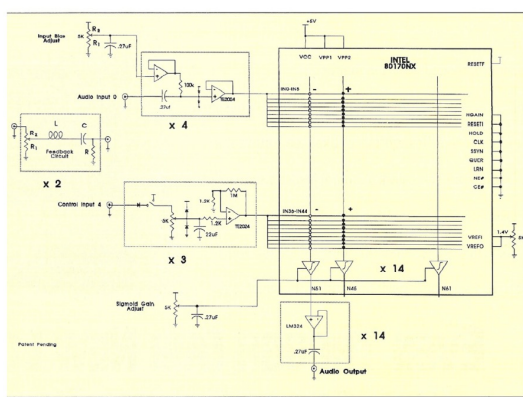
WP1b: Sonic-Social Genre: Towards Multimodal Computational Music Genre Modelling



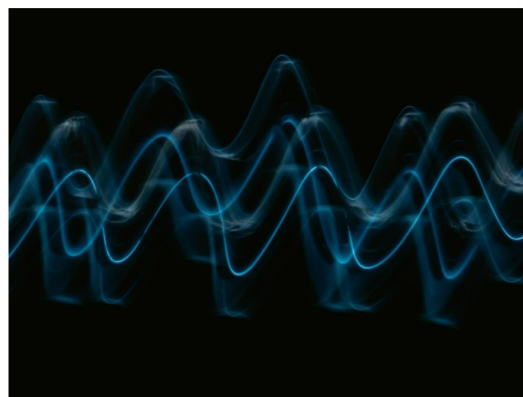
WP2a: Cultural Economies of Adaptive and Affective Music AI



WP2b: Commercial Generative Music: A Practice-Based Study of AI Music Production



WP3a: Critical Interdisciplinarity: Musician-Engineer Collaboration in Music AI Research



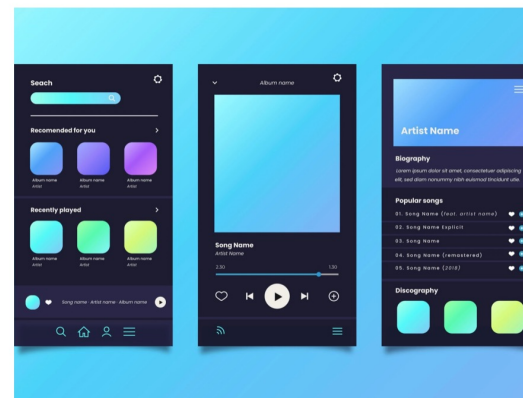
WP3b: Automating Signal Processing, Automating Aesthetic Labour and Decision-Making



WP3c: Permeable Interdisciplinary: Algorithmic Composition, Subverted



WP4a: 'Localising' Recommendation: Serving Middle Eastern Listeners and Music



WP4b: Interdisciplinary Interventions in the Design of Music Recommender Systems

MusAI research projects

Sonic-Social Genre: Towards Multimodal Computational Music Genre Modelling (Owen Green, Bob Sturm, Georgina Born)

Cultural Economies of Adaptive and Affective Music AI (Eric Drott)

Commercial Generative Music: A Practice-Based Study of AI Music Production (Oliver Bown)

Critical Interdisciplinarity: Musician-Engineer Collaboration in Music AI Research (Christopher Haworth)

Automating Signal Processing, Automating Aesthetic Labour and Decision-Making (Jonathan Sterne)

Permeable Interdisciplinary: Algorithmic Composition, Subverted (Aaron Einbond, Artemi-Maria Gioti)

'Localising' Recommendation: Serving Middle Eastern Listeners and Music (Darci Sprengel)

Algorithmic virality and West African Music and Politics (Jaana Serres)

Interdisciplinary Interventions in the Design of Music Recommender Systems (G. Born, Fernando Diaz, Andres Ferraro, Gustavo Ferreira, Jerome Ramos)

Prototyping Radically Interdisciplinary Music AI Pedagogies (Rebecca Fiebrink, Georgina Born, the team)

Topics for discussion

- Challenges and limits of **interdisciplinarity** between CS & SSHA?
- **Translating normative principles into design** of AI systems (MRS, music database)?
.. And where do we find such principles?
- **Political economy of AI beyond commercial tech**: what institutional and intellectual conditions make it possible to engage in alternative AI system design?
Public, non-profit, alternative formations?
- Cultural (and music) **curation**: doing this differently?
- From shared **exposure** (common cultural experience) to **publics**: how could this be achieved? And how measured?
- **Beyond metrics**, what other interventions in MRS would also change listening culture?

3 continuities with earlier seminars to NB:

- 1) **Interdisciplinarity**: session 1 – with SSHA-CS to fore
- 2) **Political economy** (this am) and **institutions** (Born's 4th plane of social mediation): beyond attending solely to commercial tech / Big Tech / Platform Capitalism etc; arguing for recognition of **pluralism** in political economy > agency (again) at big scale
 - Will provide 2 examples of universities, public research funding, public cultural institutions doing design differently
- 2) **'Configuring the user'** (week 1) – how the design of personalized recommender systems projects/shapes the experience of music / cultural consumption – of the 'user'

‘Measuring Commonality in Recommendation of Cultural Content to Strengthen Cultural Citizenship’

published in
ACM Transactions of Recommender Systems (2024)

Andres Ferraro, Pandora

Gustavo Ferreira, McGill

Fernando Diaz, Carnegie Mellon University

Georgina Born, University College London

Re-Engineering Recommendation – Prototyping Radical Interdisciplinary

- **Personalized recommender systems (RS)** are the dominant means of curating cultural **content**: greater automatized intervention than previous forms of curation
- Despite their personalized address, **RS have cumulative effects in shaping the wider cultures in which they're being employed (> cultural change)** – yet **current research neglects these cumulative influences both across user populations and over time**
- **Key terms here echo chambers, filter bubbles, fragmentation**
- In this light we introduce **the 'commonality' metric to measure the degree to which a RS enables a given user population to have a shared experience of specified cultural content**
- With reference to normative principles underlying public service media (PSM) systems, we identify **universality of address and content diversity in the service of enhancing cultural citizenship as key goals for RS delivering cultural content in democratic societies**
- To achieve content diversity we propose a model of **human editors accountable to larger value communities as a way of curating diverse content to promote cultural citizenship**
- We empirically compare recommendation algorithms using commonality with existing utility, diversity, novelty and fairness metrics and find **they complement existing metrics**

... 4 opening propositions

- 1) Recommender system design should move beyond a purely commercial orientation focused on personalization: should pursue alternative design paradigms guided by normative principles intended to promote the democratic development of contemporary societies as this enhances human flourishing > public good or public interest AI design
- 2) Need to acknowledge the cumulative influences of RS, which mediate wider cultural and social change, and develop ways of modifying these influences in socially progressive ways
- 3) To advance these propositions, we employ evaluation metrics as a way to incorporate alternative 'values in design' into recommender systems
- 4) We draw on research that identifies the normative principles underlying public service media (PSM) systems, principles that can then be translated into a quantitative metric
- We've attracted interest in our work from PSMs around the world – BBC, Radio Canada, Australian ABC
- BUT wider implications, not limited to PSM > aim to encourage public good/public interest work on AI design by non-profit, alternative and grassroots bodies!
- > NB: Difficult publication and conference history: IDy in CS/AI space is difficult to publish! ₇

Our method: prototyping radical interdisciplinarity

- Our research process: **sustained interdisciplinary dialogues** between 2 computer scientists (Diaz, Ferraro) and 2 scholars from SSHA with PSM expertise (Born, Ferreira)
- **Agonistic-antagonistic, sustained IDy: over a year** we instructed each other in background research, sharing ideas, deepening mutual engagement in both directions
- In this way **we translated terms from one 'side' to the other, responding to questioning about key concepts, and adapting the latter**
- Such translation across disciplinary domains can be difficult and incomplete: yet in our experience it produces hybrid thinking that generates powerful concepts and tools
- > **'Values in design' debate:** affirms this IDy translation process, cautioning that **the translation of normative principles into formalizations will always entail reductions and abstractions** [> Fabio yesterday]. **The question is: which reductions and abstractions should be tolerated in order to pursue core normative dimensions?**

Phil Agre, 'Toward a critical technical practice' (1997): a manifesto for 'the reflexive work of critique'

- 'As an AI practitioner..., I had incorporated the field's taste for technical formalization so thoroughly into my own cognitive style that I literally could not read the literatures of non-technical fields... The problem was... I insisted on reading everything as a narration of the workings of a mechanism'
- 'This experience led me into full-blown dissidence within the field of AI. My first breakthrough came when... it finally occurred to me to stop translating [other] disciplinary languages into technical schemata, and learn them on their own terms. This was difficult because my training had instilled in me... a single mission: transforming vagueness into precision through formalization'
- 'AI, like any other field, ought to have a space for critical reflection on its methods and concepts... [which would] encourage connections with methods and concepts from other fields'
- 'AI's construction of itself as a self-contained technical discipline... [is] a powerful force for intellectual conservatism. Critics will be asked, "what's your alternative?", within a tacit system of discursive rules that rules out alternatives from the start... AI's intricate, largely unconscious cultural system ensures that all innovations... [are] enmeshed in traditional assumptions and practices'
- 'Critical practice is essential to make sense of such things, but its goal should be complex engagement... A critical technical practice will... require a split identity -- one foot planted in the craft work of design and the other foot planted in the reflexive work of critique'

For a plural political econ of media/AI – beyond commercial tech

- **Plurality in political economy of media (and music) a longstanding reality – and a core theme of media and popular music studies > more complex than US/global oligopolies**
- Neglected by US-centric critical AI literature – and US lacks strong (national) public media!
- **What plurality? NB Existence of both:**
- **1) Publicly (often) funded, national media institutions and systems regulated for public service/ interest principles** (PSMs in UK, Europe, Canada, Asia, Australia, L. America, Africa...) – distinct from government (authoritarian) media (eg Russia) via (imperfect) institutional and political independence, accurate news, democratic mandate, commitment to diversity > **idea of publicly-funded platform/distributor alternatives to dominant commercial platforms**
- PSM envisaged, from early C20th, as means of **intervening in media markets to set and uphold standards** of free speech, accuracy in news, democratize access to culture etc
- **2) Independent, alternative, grassroots, pirate labels / distributors / media** – small scale, ‘underground’, ‘radical’ etc... ‘petty’ capitalism, or anti-capitalist: still uphold plurality – BUT now mediated by gatekeeper platforms/distributors (discussion earlier)
- **These institutions matter: source of innovation, talent, public sphere, natl imaginaries!**

Normative principles:

From PSM (BBC, C4 etc in law): universality and diversity...

From political theory: cultural citizenship...

- **Concept of cultural citizenship (CC)** foundational for recent democratic theory: a striking feature of recent political discourse 'is the increasing confluence of culture and citizenship'
- Recognises **profound social transformations wrought by globalization, migration, decolonisation, the growing heterogeneity of nation states, and the intensification of identity politics among subaltern and marginalized groups**
- Given these profound changes, **CC draws attention to a new domain of cultural rights: the right to symbolic presence, dignifying representation, and the maintenance of distinct cultural identities** – where such cultural pluralism 'enriches... the fabric of society' (Isin)
- To promote CC, PSM and other democratic and public interest channels of cultural production and distribution have a responsibility to curate a plurality of content in order to stimulate *intercultural* and *intracultural* exposure and dialogue, interest in/empathy for cultural diversity
- **Music has always been particularly significant as a medium for experiencing cultural diversity – and thereby acts as a powerful means of enhancing cultural citizenship!**

Born, 'Mediating the public sphere' (2012) and 'Digitising democracy' (2005)

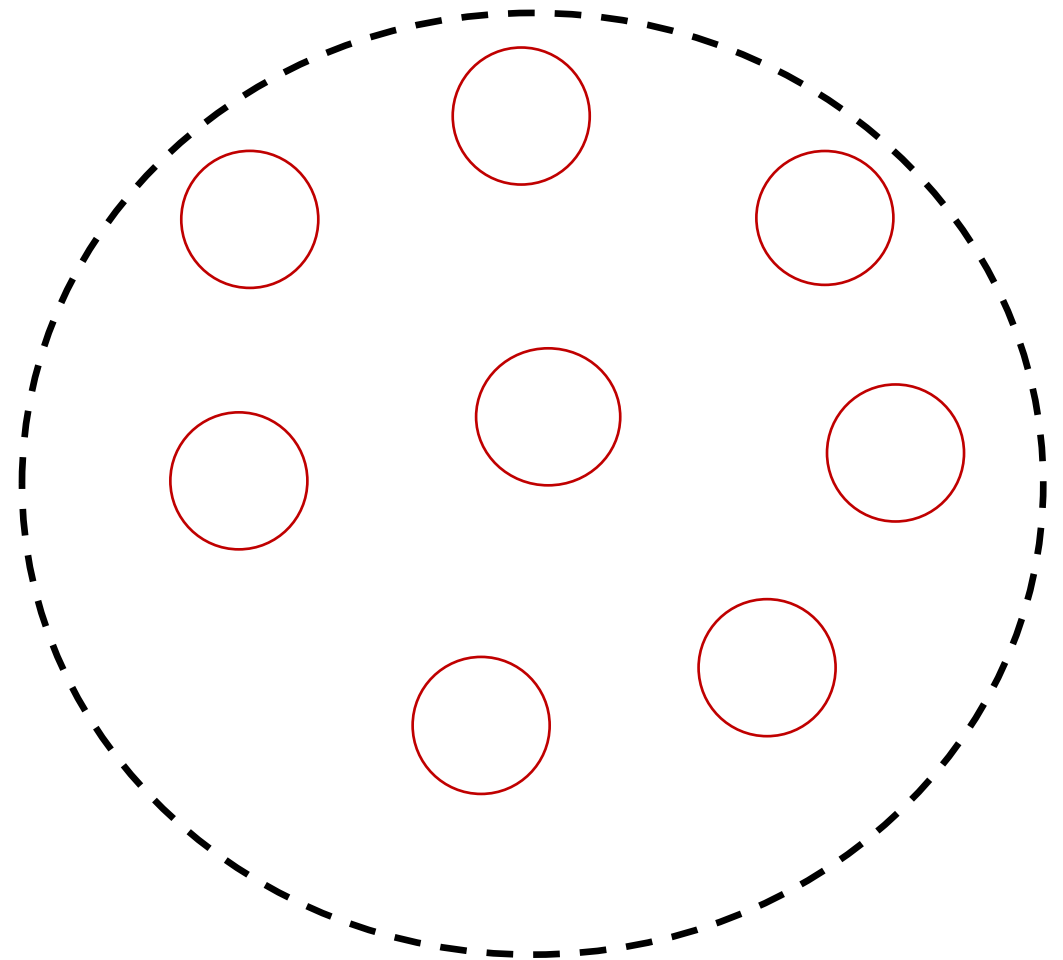
Why universality (or commonality) of address?: On the public-making significance of shared cultural experiences

- **PSM principle of universality:** need for content that reflects and caters to cultural and social diversity of cosmopolitan societies, while enhancing social cohesion via creation of 'common culture'
- **Stuart Hall:** 'The quality of life for black or ethnic minorities depends on the whole society knowing more about the "black experience"' through black cultural production – music, film, literature, art
- **Bikhu Parekh, James Tully, Seyla Benhabib:** mutual cultural recognition and the expansion of cultural referents essential to the wellbeing of pluralist societies, but doesn't obviate need for integration via fostering of common experience and common identities, enabling hybridity across social boundaries
- **Nancy Fraser:** existence of plural counter/subaltern public-spheres must be accompanied by 'a more comprehensive arena in which members of different publics talk across lines of cultural diversity'
- **Michael Warner:** shared cultural experiences generate *publicness* – experience in common with others, with whom this can be enjoyed and debated: a common social imaginary
- Such effects are attenuated by RS personalization! > fragmentation, echo chambers, filter bubbles
- Our commonality metric helps to (re)create universality of address in RS content curation

Theorists of publics respond to their times: 1990s-2000s: (feminist) democratic political theory

‘Subaltern counter-publics’

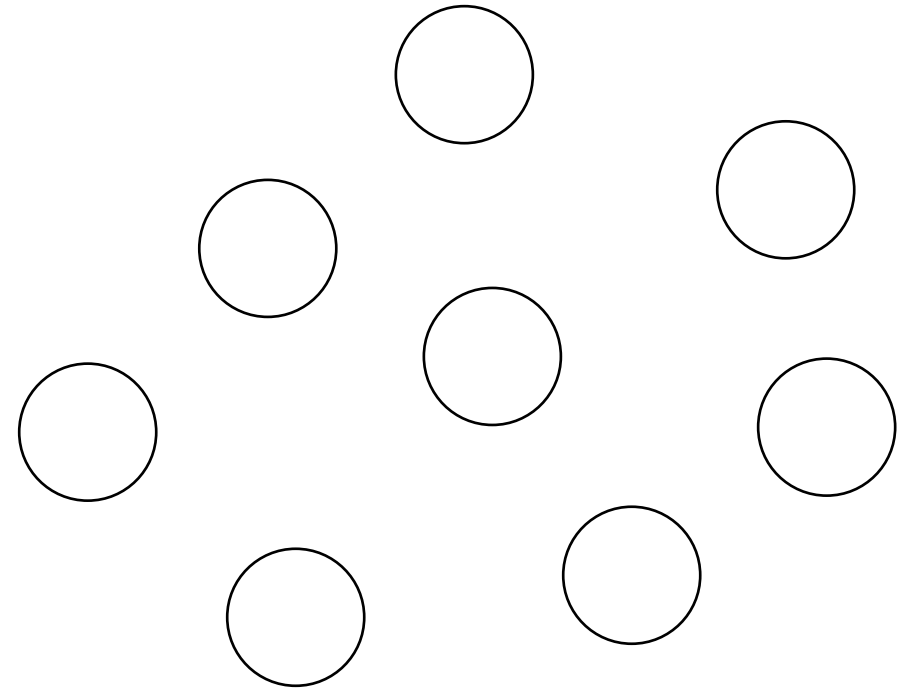
Before social media: influence of feminism, multiculturalism, identity politics: against the suppression of difference by universalist (‘mass’) publics > advocate pluralist ‘subaltern counter-publics’, forming a nexus of multiple publics (Nancy Fraser)



By 2010s: from counter-publics to echo chambers!

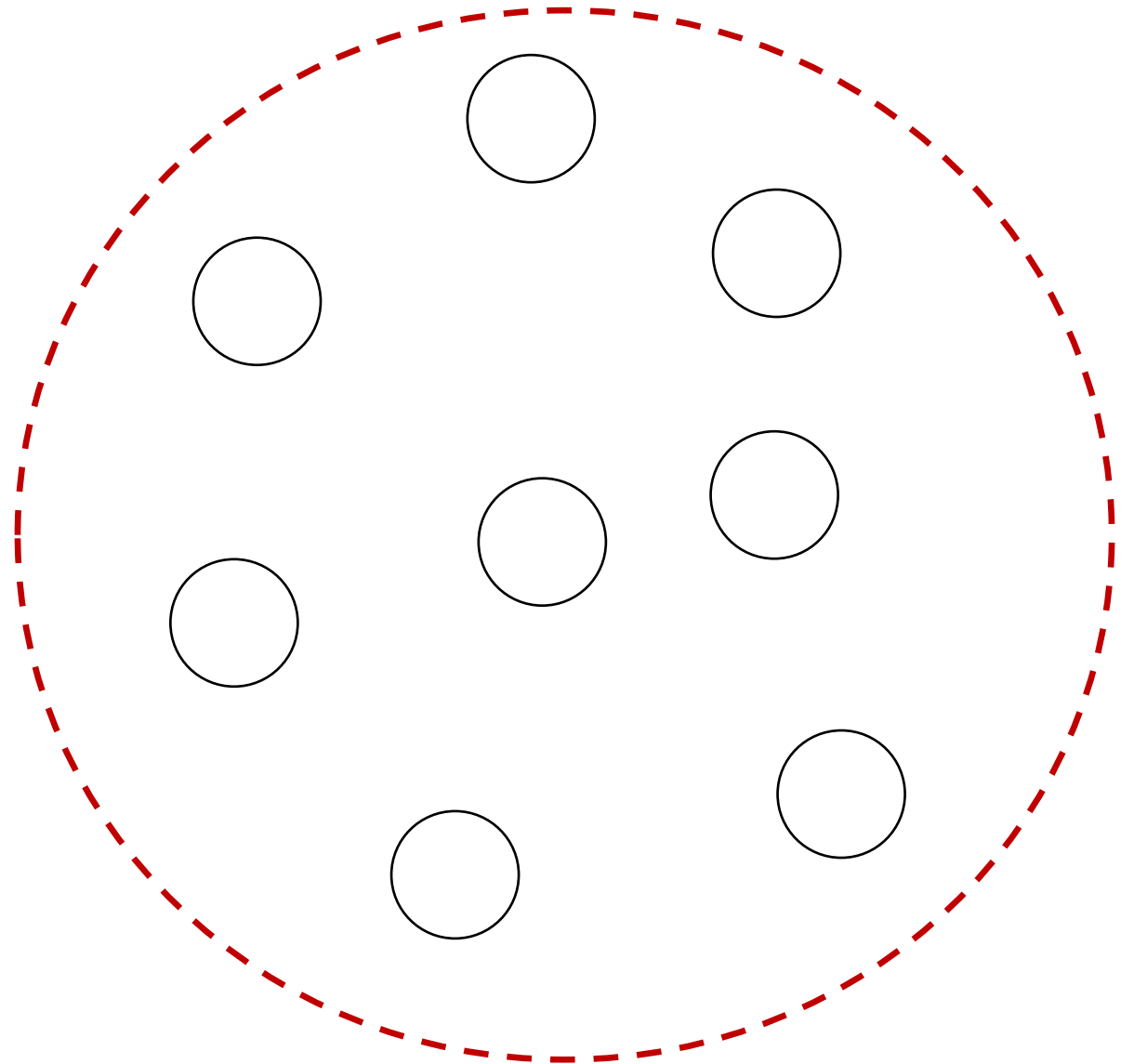
Echo chambers, filter bubbles

Rise of social media and
personalization >
fuels concerns about
excessive fragmentation +
polarization >
causing a loss of
universalistic publics, shared
experience



Now: post-personalization?

Our proposal: need to recreate and reinforce **common, shared experiences in guise of 'universalistic' publics** (commonality) to stimulate socially-shared information, knowledge, culture, imaginaries... **BUT this is not about consensus!:** but a common public space in which democratic deliberation, disagreement, agonism, can proceed (**Chantal Mouffe**)



Lessons? Dialectic between mediated universality & diversity

- Is this harder in digital present? Via the **individuating (siloing) effects of personalisation?**
- NB: links between rich communicative democracy and wellbeing of national political culture
- ... in contrast to social media's and the US media's 'logic of segmentation' (Gandy, Katz)
- Need *both* **services catering for and reflecting diverse interests of social and cultural minorities**
- ... And **services that enhance social unity and cohesion** via creation of a 'common culture' and provision of common information and entertainment
- **Born 'Digitizing democracy' (2013):** how PSM can enhance such rich communicative democratic spaces:
- > **3 dimensions of universality/diversity dialectic**, of encounter between majority & minorities:
 - 1) **Universality**: services / platforms / modes of address that **provide common experiences in news, entertainment, education** – while hosting diverse and divergent (minority) perspectives
 - 2) **Inter-cultural communication**: enabling minority to speak to majority & other minorities
 - 3) **Intra-cultural communication**: (niche) services enabling minority to speak (mainly) to itself

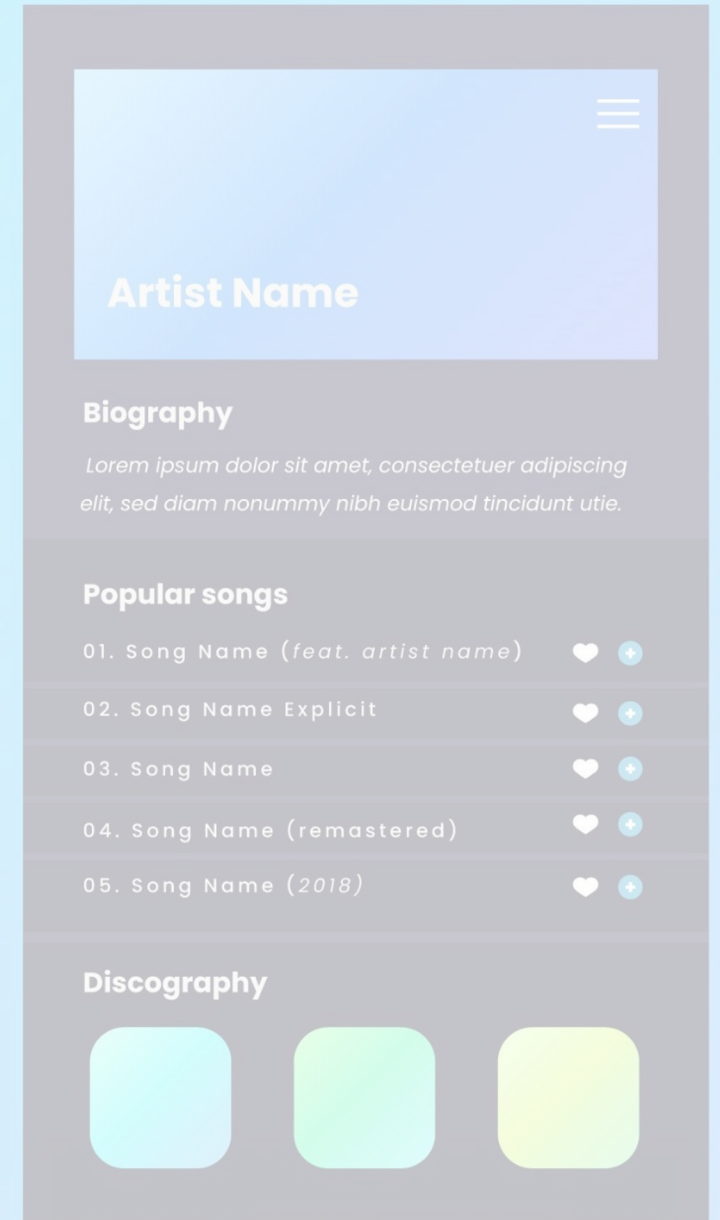
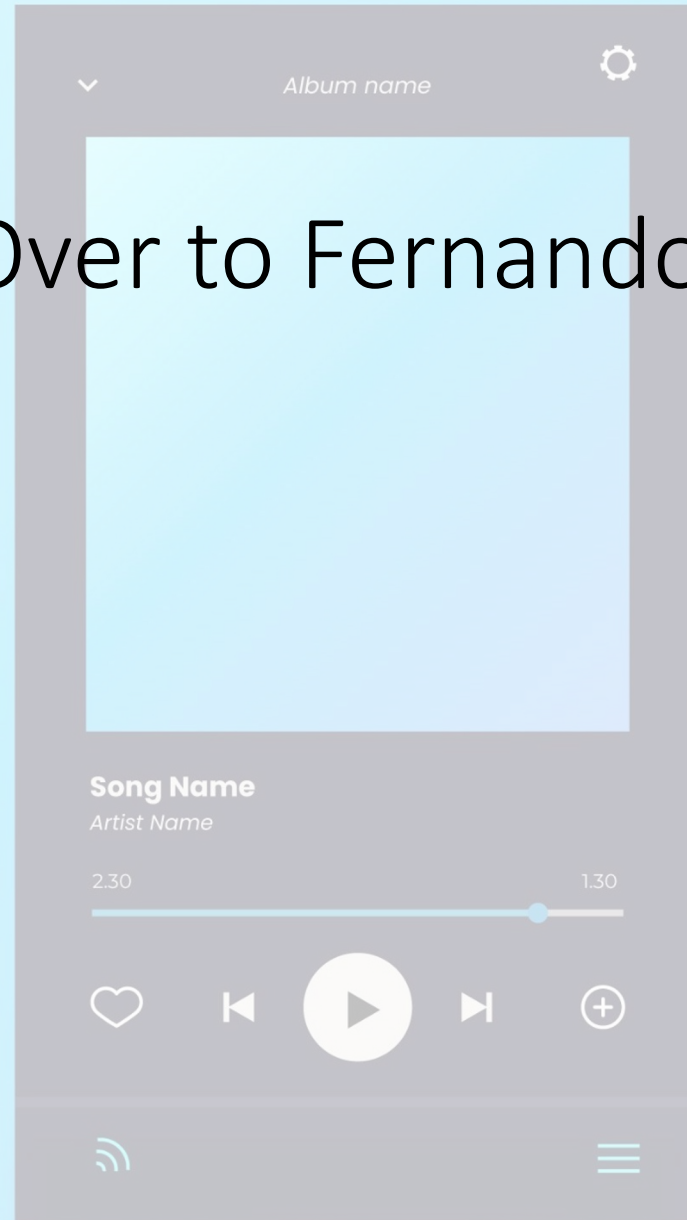
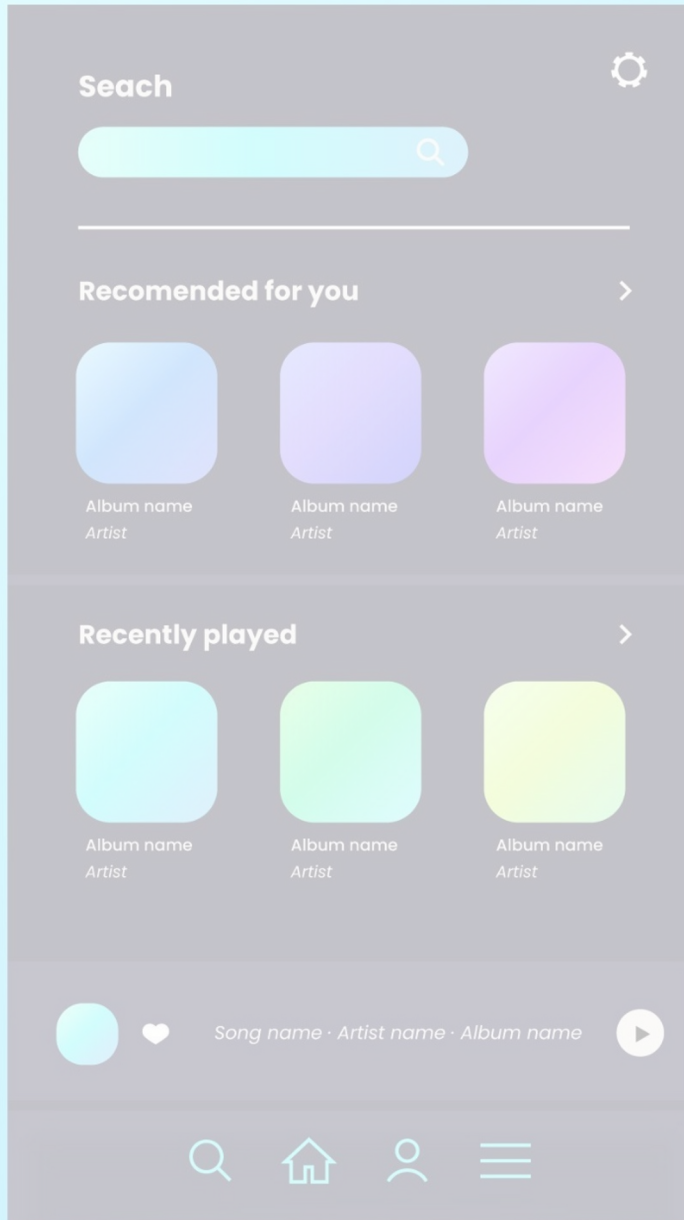
And how do we conceptualise diversity?

- If diversity of shared cultural experience is a means of enhancing cultural citizenship, **how should diversity in RS** be conceptualised towards this end?
- **We're concerned not only with individualization but with the need to boost diversity in the curation of cultural content and benefits of promoting shared diversity of cultural experience**
- 2 most obvious vectors of diversity are **content diversity** (eg in terms of range of genres, and within genres) and **diversity of source or producer** – where greater diversity of producer is likely to favour (but doesn't guarantee) greater diversity of content
- The various biases shown by the under-representation of cultural content with respect to gender, race, class and region (ie diversity of source) correspond to wider core-periphery dynamics and geographical inequalities in the cultural industries: **RS tend to mirror these inequalities, favouring Western popular content, in the English language, from major producers**
- Increasing **diversity of source**, as an issue of equity and as it bears on diversity of content, is therefore crucial for RS oriented to enhancing cultural citizenship
- We test commonality with **3 categories of under-represented source in music, film and literature: female & non-binary producers; independent production; and non-Western sources**

On human editors and ‘value communities’ in RS curation: avoiding essentialism, gaining insider (community) insight

- RS are **assemblages or ‘sociotechnical systems’** >
- **To avoid essentialism and reification in classificatory judgments** of content and source diversity, we advocate **human editorial processes that draw on communally-validated classifications that respect evolving subtleties both of demographic categories and of genres**
- **So the role of human editors is to reflect on the diversity of a RS with respect to given categories** by drawing on insights generated by a larger ‘value community’ knowledgeable about relevant cultural expressions, their demographic basis and genre designation
- **> A sociologically-normative model of judgement:** by ‘value community’ we refer to communities generated by shared cultural passions and interests (among them genre communities), who embody an evolving consensus about these interests and their relationship to categories of social identity, and about which members have (varying degrees of) expertise
- **The human editors act as conduits for these larger communities of interest and judgement, and their judgments are legitimised and validated in turn by this relationship**
- We aim to **redesign accountable & legitimate editorial processes in curation of diversity**

Over to Fernando...



Recommender systems

From: Hal Varian <hal@alfred.sims.berkeley.edu>
Date: Mon, 18 Mar 96 09:25:50 -0800
Subject: "collaborative filtering" name game---how about "recommenders"?

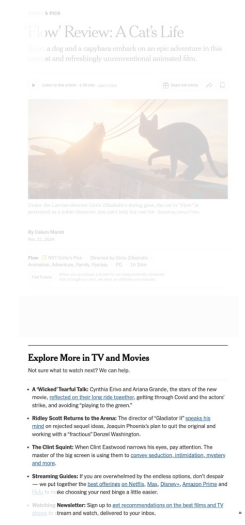
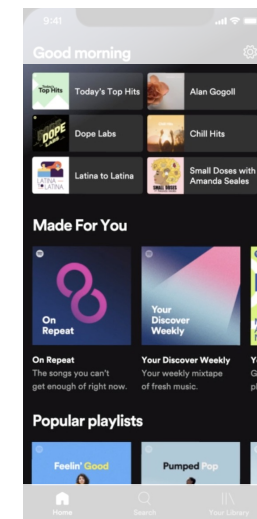
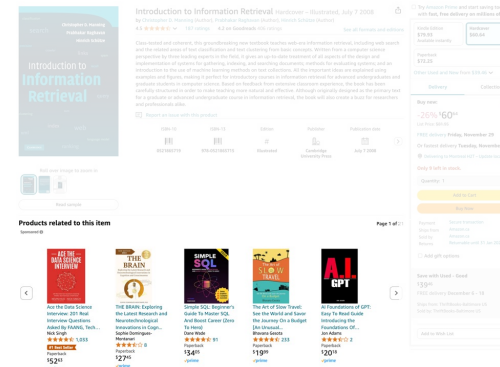
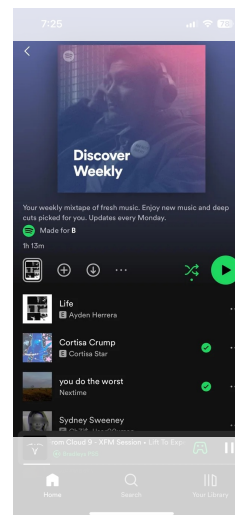
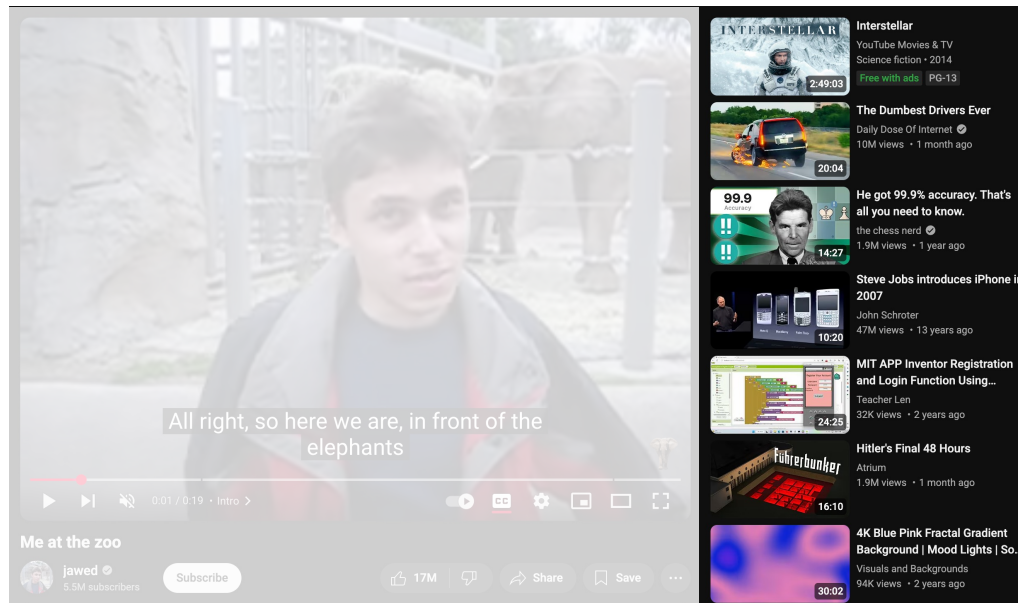
I'm going to take up the "Resnick challenge" and try to come up with a better name than "collaborative filtering". My suggestion is that the general category of interest is that of "recommenders". In this category we can consider several subcategories:

recommender agents
community recommenders
expert recommenders
etc.

Hal Varian, Dean
SIMS,
University of California
Berkeley, CA 94720-4600

- Relatively young field emerging from collaborative filtering.
- Methods and evaluation based on concepts from information retrieval, the field underlying search engines.
- Task: *given some information (e.g., user behavior, query), sort a catalog of items in decreasing order of relevance.*

Recommender systems



Recommender systems

recommend some music to listen to while I make guacamole.

Making guacamole calls for something upbeat, relaxed, and rhythmic enough to keep you chopping in time.

Sunny, feel-good mix

- [Khruangbin](#) — especially the album [The Universe Smiles Upon You](#)
- [Santana](#) — try the song [Oye Como Va](#)
- [Buena Vista Social Club](#) — the song [Chan Chan](#)
- [Bomba Estéreo](#) — [To My Love](#)
- [Calexico](#) — desert-sunset vibes

If you want a modern Latin groove

- [Natalia Lafourcade](#) — [Hasta la Raíz](#)
- [Monsieur Periné](#) — [Nuestra Canción](#)
- [Jorge Drexler](#) — mellow and sophisticated
- [Vicente García](#) — warm acoustic grooves

If you're hosting people in 20 minutes

Put these on in order:

1. [Chan Chan](#)
2. [Oye Como Va](#)
3. [Hasta la Raíz](#)
4. [To My Love](#)
5. [Nuestra Canción](#)

That sequence goes from "casually mashing avocados" to "everyone is hovering around the bowl before it's finished."

If you'd rather lean indie, jazz, electronic, classic rock, or something more Mexican-specific, I can build a more focused guacamole soundtrack. 🥑🎶

recommend some music to listen to while I make guacamole.

Making guacamole calls for something that sets a chill, upbeat vibe—here are some great options:

Classic choices:

- [Sublime](#) - reggae-infused vibes perfect for relaxed prep work
- [Slightly Stoopid](#) - laid-back ska and reggae that keeps things breezy
- [The Incredible String Band](#) - folksy and groovy, great for focused cooking

Modern feel-good:

- [Glass Animals](#) - smooth electronic-pop that's both creative and easy-listening
- [Tame Impala](#) - psychedelic and mellow, ideal for creative work
- [Khruangbin](#) - instrumental funk-groove that's hypnotic without demanding attention

Upbeat options:

- [The Black Keys](#) - bluesy rock that's energetic but not overwhelming
- [Anderson .Paak](#) - funky, soulful hip-hop with great rhythm
- [Alt-J](#) - quirky indie with fun instrumentation

Playlists to consider:

- Spotify's "Chill Vibes" or "Good as Hell" playlists
- YouTube Music's "Relaxing Indie" or "Feel Good Summer"
- Apple Music's "A-List Independent"

My top pick: Go with [Khruangbin](#) or [Glass Animals](#) — they're engaging enough to keep you energized while chopping and mixing, but chill enough that they won't make you feel rushed. Plus, there's something about their grooves that just feels right while assembling fresh ingredients.

What kind of mood are you leaning toward—more relaxed and mellow, or something with more energy?

recommend some music to listen to while I make guacamole.

Making guacamole is all about the rhythm—mashing the avocados, finely dicing the jalapeños, and squeezing the lime. You need a soundtrack that feels sunny, vibrant, and rhythmic without being so intense that you accidentally lose focus with the chef's knife.

Here are three distinct directions depending on the exact vibe you want in your kitchen:

Vibe 1: The Sunny Backyard Party (Traditional & Neo-Cumbia)

Perfect for a classic, high-energy prep session. The rhythms practically match the texture of a molcajete.

- ["Cumbia Sobre el Mar"](#) - Quantic & Flowering Inferno
(A gorgeous, horn-heavy, dub-infused cumbia that makes the kitchen feel instantly warmer.)
- ["La Colegiala"](#) - The Boy Next Door x Fresh Coast ft. Jody Bernal
(A modern, bouncy take on a classic rhythm that keeps your energy up while dicing onions.)
- ["Hasta la Raíz"](#) - Natalia Lafourcade
(Beautiful, acoustic-driven Mexican indie pop that provides a soulful, rhythmic background.)

Vibe 2: Tropicália & Bossa Groove (Smooth & Sunny)

If you want something breezy, laid-back, and effortlessly cool while you gently fold in the cilantro.

- ["Mas Que Nada"](#) - Jorge Ben Jor
(The ultimate Brazilian groove. It is physically impossible to be in a bad mood while listening to this.)
- ["Taj Mahal"](#) - Jorge Ben Jor
(An infectious, driving rhythm that keeps a steady pace for mashing those avocados.)
- ["Flor de Lis"](#) - Djavan
(Smooth, upbeat samba jazz that pairs perfectly with a crisp drink.)

Vibe 3: Desert Rock & Funky Psych (Spicy & Edgy)

If your guacamole has extra jalapeños and you want a gritty, desert-inspired, guitar-heavy soundtrack.

- ["Connaissais de Face"](#) - Khruangbin
(Laid-back, bass-heavy psychedelic rock. It creates a smooth, hypnotic groove perfect for focused kitchen work.)
- ["Como Me Queres"](#) - Chicano Batman
(Smooth, retro-soul Latin psych-rock with soaring organs and a great rhythm section.)
- ["Malamente"](#) - Rosalía
(Modern flamenco-pop fusion with sharp claps and a driving beat to keep you moving.)

Pro Tip: Don't over-mash! Keep some small chunks of avocado intact for a better texture, and always add the salt to your aromatics (onion, chili, cilantro) *before* mixing them into the avocado to draw out the essential oils.

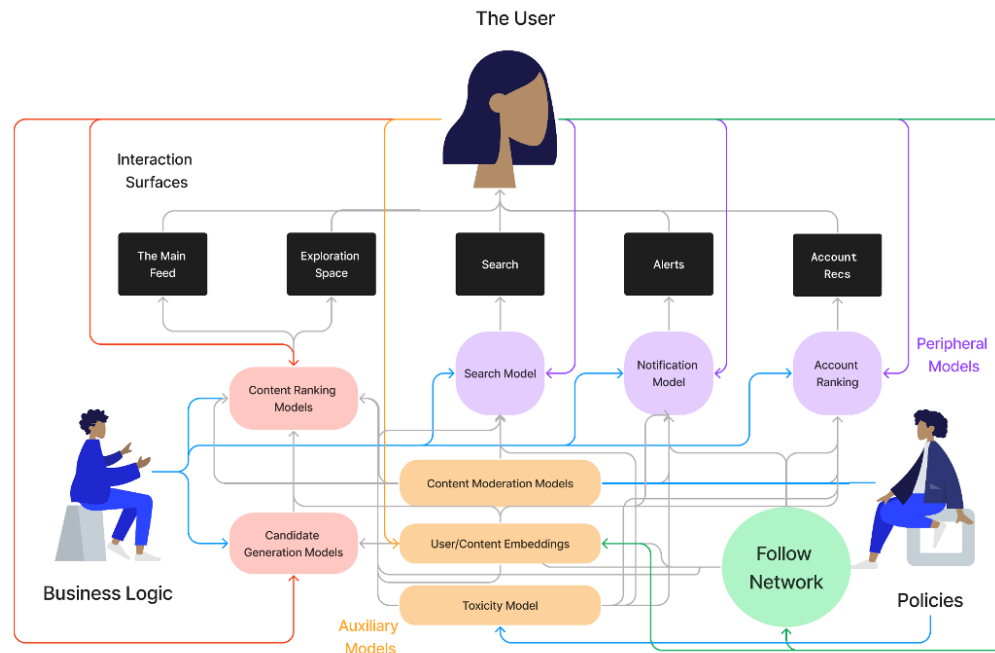
Need a rock-solid, authentic recipe to go with the music?

Recommender systems

- Algorithms underlie,
 - platforms (e.g., music, video, news, audiobooks, games)
 - tools (e.g., driving directions, predictive typing)
- Sometimes recognizable, but often embedded in larger systems.

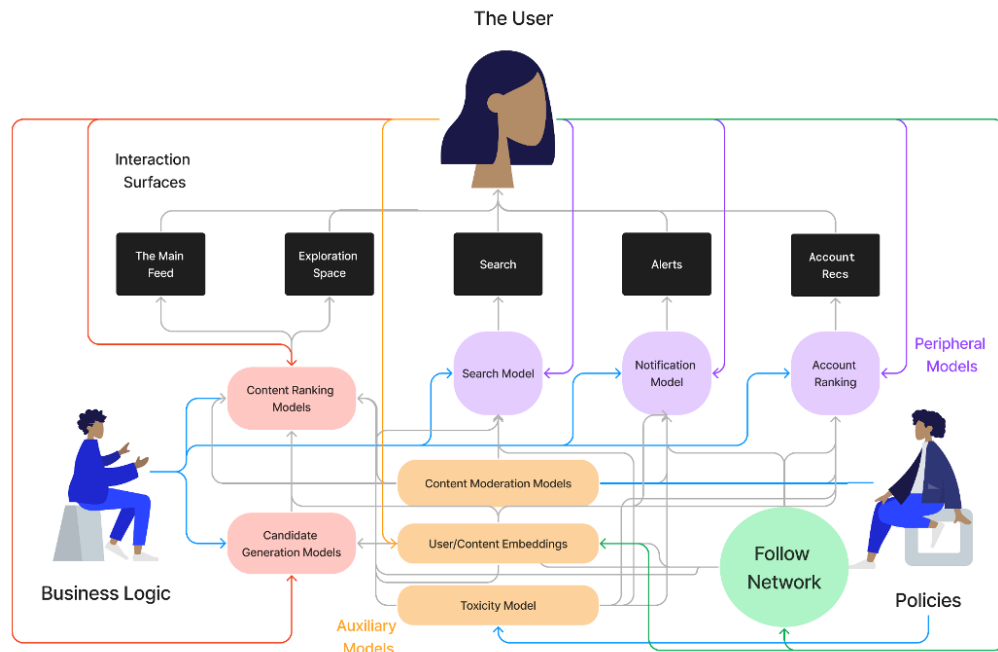
Recommender systems

- Even when recognizable, recommender systems are composed of an interconnected set of models, technologies, and people.



Kristian Lum and Tomo Lazovich, "The Myth of The Algorithm: A System-Level View of Algorithmic Amplification", 2023.

Points of intervention



Kristian Lum and Tomo Lazovich, "The Myth of The Algorithm: A System-Level View of Algorithmic Amplification", 2023.

- Even when recognizable, recommender systems are composed of an interconnected set of models, technologies, and people.
- In order to design recommender systems that are aligned with PSM principles, we can/should intervene in multiple parts of the technology.

Points of intervention

catalog
policy
interface
algorithm
data annotation
metrics

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- In order to design recommender systems that are aligned with PSM principles, we can/should intervene in multiple parts of the technology.

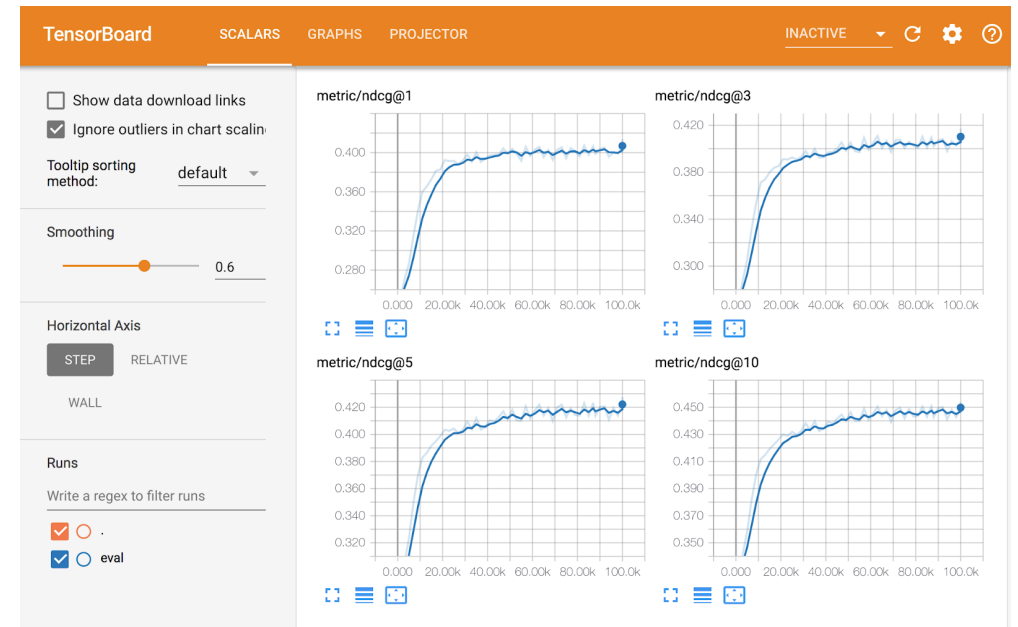
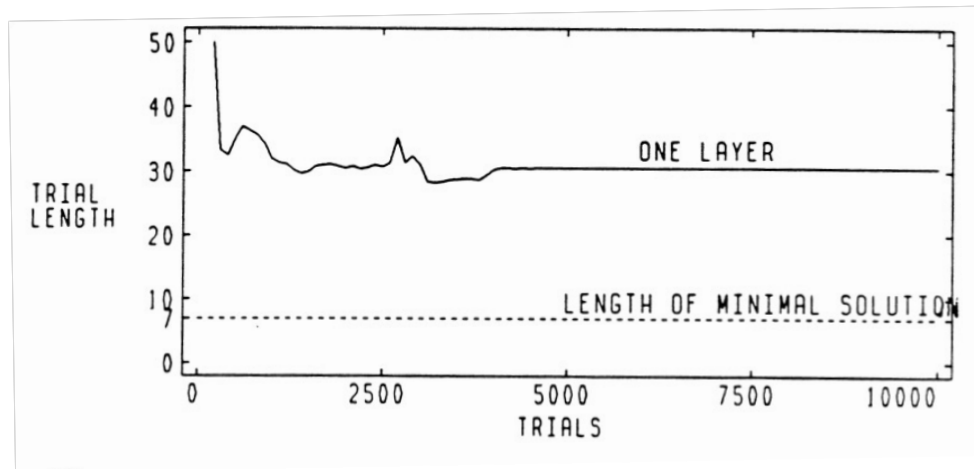
Evaluation metrics

- **evaluation metrics** or **key performance indicators** are quantitative measures of some property of the system, usually quality.
 - can be used as a heuristic to pick algorithms (i.e. answer “what is better?”)

property	metric
utility	“how many users preferred system output A to B in a laboratory setting?”
utility	“how many users engaged with system output A to B in a production setting?”
misinformation	“is this system output factual?”
diversity	“does this system output reflect all perspectives?”

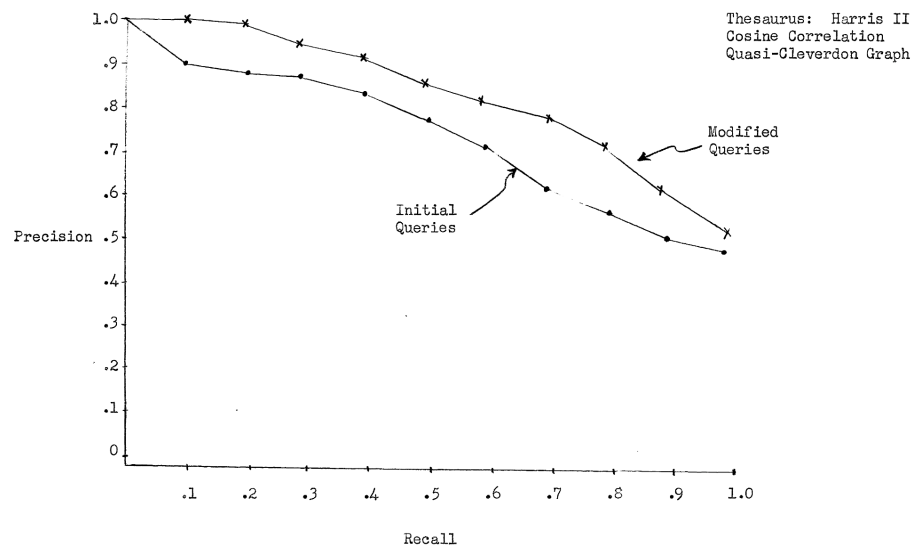
Metric use: system design

- can be directly optimized by modern machine learning algorithms.
 - “prefer a model if it has a higher metric”



Metric use: research progress

- determining state of the art in the machine learning research community
 - “accept papers based on measured quality”



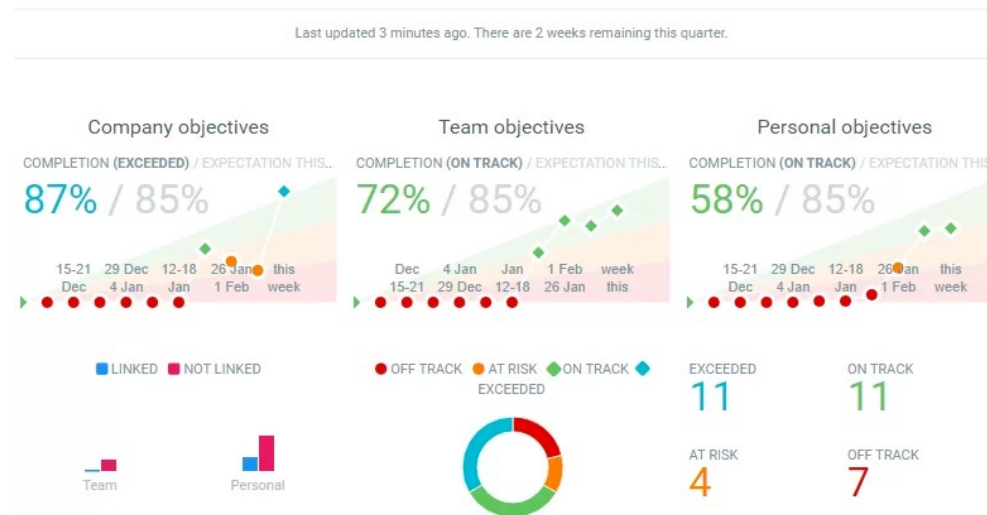
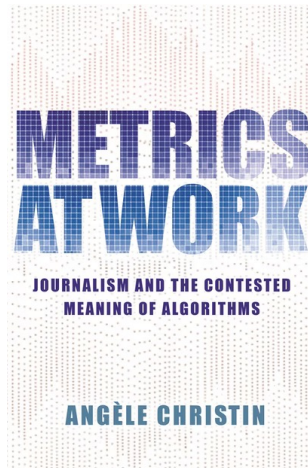
Long Answer Leaderboard										
Rank	Model	Participant	Affiliation	Attempt Date	F1	Precision	Recall	R@P = 90	R@P = 75	R@P = 50
1	PoolingFormer	PoolingFormer_Team	MSRA&Dynamics 365 AI	1/13/21	0.79823	0.7847	0.81224	0.62448	0.8379	0.88748
2	ClusterFormer	Dynamics_365_AI_Research	Microsoft	9/28/20	0.77969	0.78471	0.77473	0.56986	0.80654	0.86686
3	ETC-large	philly_pham	Google Research	5/24/20	0.7778	0.77476	0.78087	0.52204	0.79864	0.86598
4	longformer_modified	Zhixiaobao	Ant Group	12/6/21	0.77295	0.77758	0.76837	0.51941	0.78921	0.86225
5	ReflectionNet-ensemble	Wide_Field	Microsoft STCA NLP Group	2/9/20	0.77185	0.76791	0.77583	0.53345	0.78526	0.85238

Recommendation Policy	β	% Loss in Fairness	% Loss in Relevance	% Gain in Satisfaction
Only Fairness	N/A	0	57.7	-35.3
Only Relevant	N/A	69.1	0	0
Interpolated	0.5	3.32	48.7	-32.2
	0.7	42.7	9.8	-10.2
	0.9	64.7	0.06	-1.1
probPolicy	0.5	16.3	44.8	-29.0
	0.7	30.8	32.7	-20.6
	0.9	48.2	17.6	-11.1
GuaranteedR	0.5	37.7	19.6	-14.2
	0.7	51.7	7.8	4.4
	0.9	63.9	0.59	22.1
Adaptive - I	N/A	17	20.2	9.0
Adaptive - II	N/A	15	21.2	12.1

Table 3: Comparing loss in fairness & relevance, with gains in satisfaction for different recommendation policies.

Metric use: organizational evaluation

- making decisions in companies
 - “deploy models that improve a metric”
- assessing contributions of teams and individuals in companies
 - “reward employees and teams if they improve a metric”



THE LEADING QUESTION

What does creating strategy with AI involve?

*Organizations must first realize that their key performance indicator (KPI) portfolio represents their strategy.

*They can then use machine learning applications to choose, measure, and optimize their KPIs.

*They must manage their data as an asset in order to enhance their KPIs and help their machines learn.

Metrics in recommender systems

- *online metrics* aim to measure the quality of a live system.
 - presented on dashboards for continuous monitoring
 - based on real user behaviors (e.g., clicks, streams, shares)
 - designing a system *only* based on online metrics can be risky
 - slow to detect changes in quality
 - puts real users at risk during experiments
- *offline metrics* aim to measure the quality in simulation.
 - based on statistical models of user behavior and labeled data
 - always an approximation of real user behavior (an approximation of quality)
 - allows for faster iteration on model development under safer conditions

Structure of offline metrics

$$\sum_{i=1}^n \Pr(i) R(\pi_i)$$

mathematical model



labeled data

- a mathematical model of behavior
 - how does a user examine a ranking of items?
 - how does a user {click, stream, consume} individual items?
- data to support modeling
 - labeled data from annotators (e.g., relevance of items to specific users)

People in offline metrics

$$\sum_{i=1}^n \Pr(i) R(\pi_i)$$

mathematical model



labeled data

- a mathematical model of behavior
 - functional form of the model
 - parameters of the model
- data to support modeling
 - labeling instructions
 - interpretation of instructions

Measuring commonality

- **core question:** can we integrate the principles from PSM into existing ways of measuring performance?
- **caveats**
 - metric design is iterative; this is a draft
 - metrics carry strong assumptions which we seek to iteratively relax
- in order to measure commonality of a system, we need to quantify,
 - the type of common experience
 - the degree of shared common experience

Measuring commonality

- Type of common experience
 - categories determined by human editors who are accountable to larger value communities
 - grounds common experience in a specific type of diversity
- Degree of shared common experience
 - can use statistical models of *how* people examine recommendations to measure the probability that all users are exposed to the selected content
 - relies on *everyone* sharing the same experience (not just some or most)

Relationship to other metrics

- existing measures of diversity in recommendations focus on,
 - diversity toward the end of revenue (e.g., retaining users, reducing cost)
 - diversity within a specific users' recommendations (e.g., not concerned with a *shared* experience)
- existing measures of fairness in recommendations are similarly limited

Empirical behavior

- analyzed the behavior of the commonality metric on multiple public datasets covering music, movies, books
- empirical results demonstrate good metric properties,
 - high convergent validity: commonality is correlated with diversity and fairness metrics
 - high discriminant validity: commonality is not correlated with personalization-focused utility metrics and provides complementary information
 - good robustness to label noise (e.g., annotation errors)

Moving forward

- translating commonality to online evaluation
 - ongoing conversations with existing PSM organizations
- validating commonality with longer term user behavior
 - does improving commonality affect how users consume content?
- exploring interventions beyond measurement
 - metrics are not enough
 - what changes can/should be made to e.g., the catalog, interface, algorithms?



Discover Podcasts



Stalked

A digital intruder is watching Hannah, threatening her, even impersonating her.



Americast

Weekly insight and analysis on what's happening inside President Trump's White House.



Newscast

Au Revoir Marine Le Pen?



Good Bad Billionaire

How did the richest people on the planet make their billions?



Diddy On Trial

The shocking allegations and conspiracies surrounding hip hop mogul Sean 'Diddy' Combs



Sliced Bread

Sam White looks at profitable everyday products and future ways we might spend our Dough.

Pinned
programme

Through the Map we can see different types of people

Metropolitan Professionals

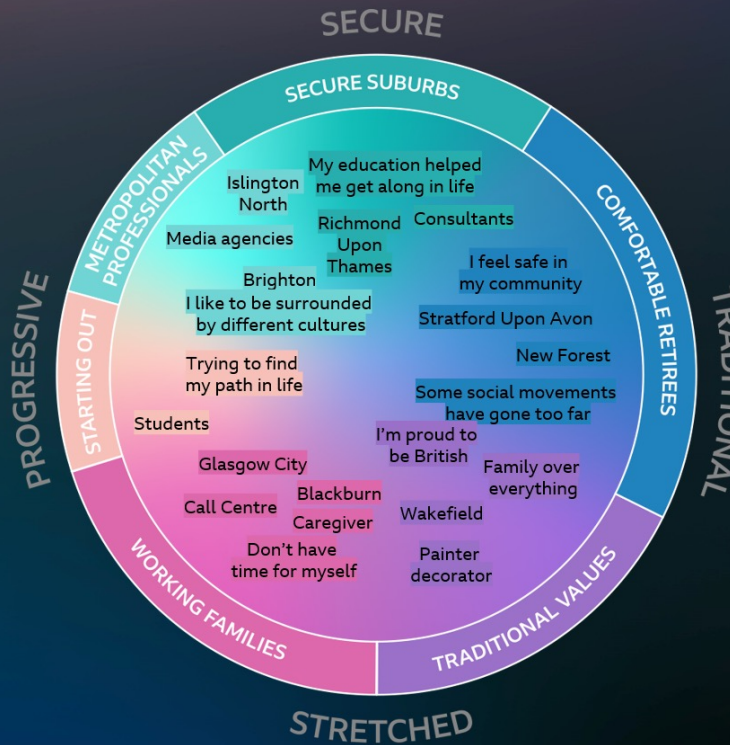
Younger (25-44yrs), university educated, and living in big city affluent areas.

Starting Out

Youngest (mostly all U25), studying or starting out in their first jobs, mainly urban areas.

Working Families

25-45, many with families, in more socio-economically deprived areas away from London and the South East.



Secure Suburbs

35-54 some have kids who are older and may have left home, living in urban areas around London, South and West

Comfortable Retirees

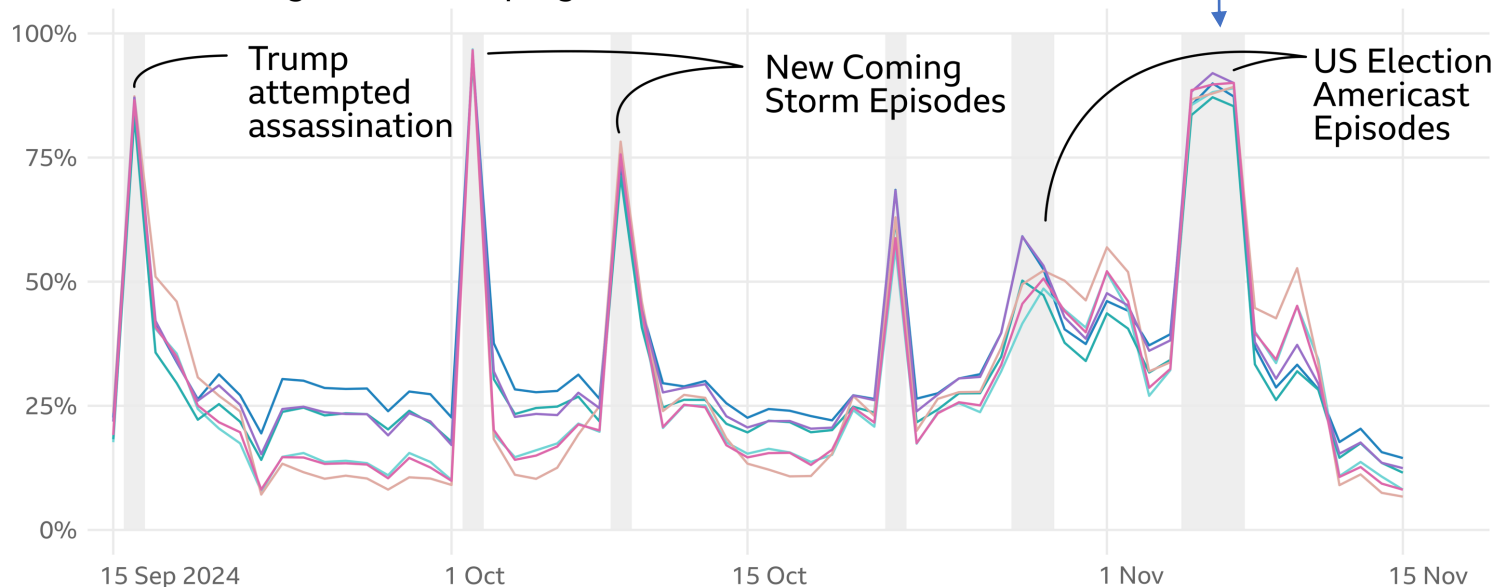
Our oldest group, 55+ most of these people are retirement age living in and around the UK's countryside

Traditional Values

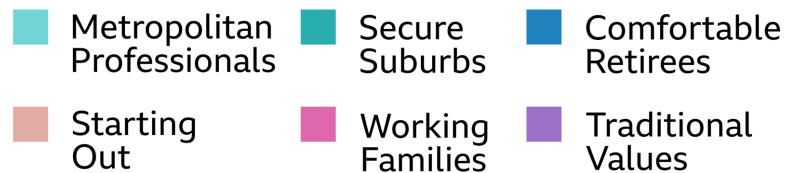
Older families (45+) and empty nesters living in post-industrial towns in North of England and Scotland

Americast & The Coming Storm

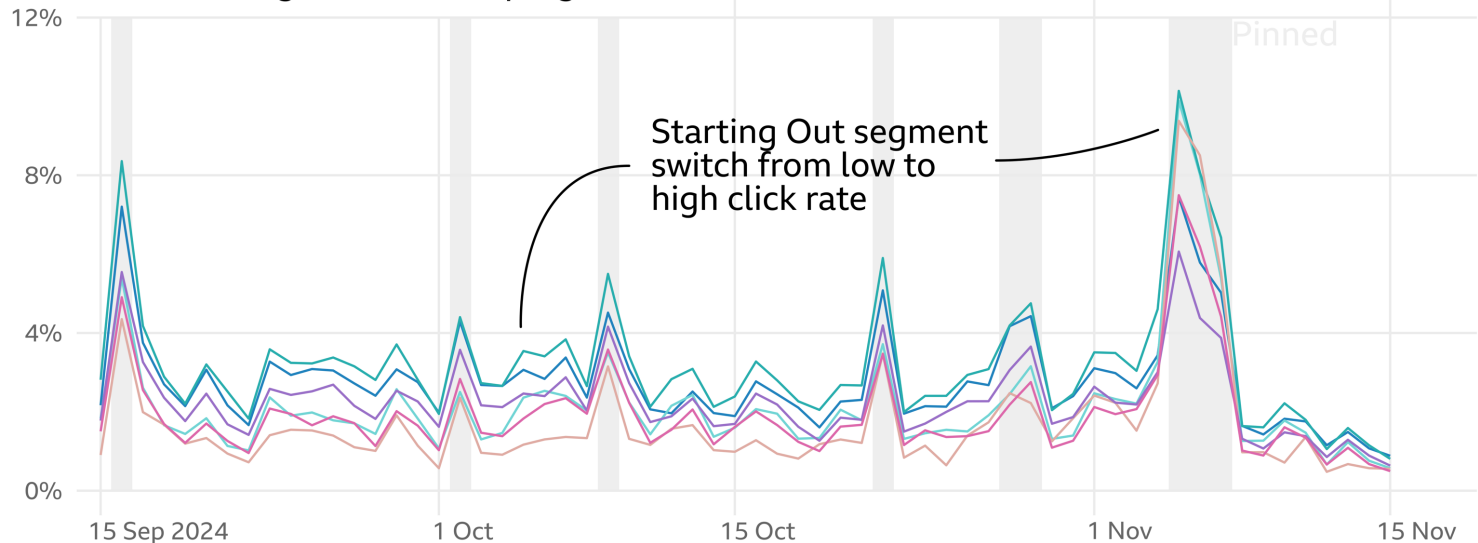
% users in segment shown programme



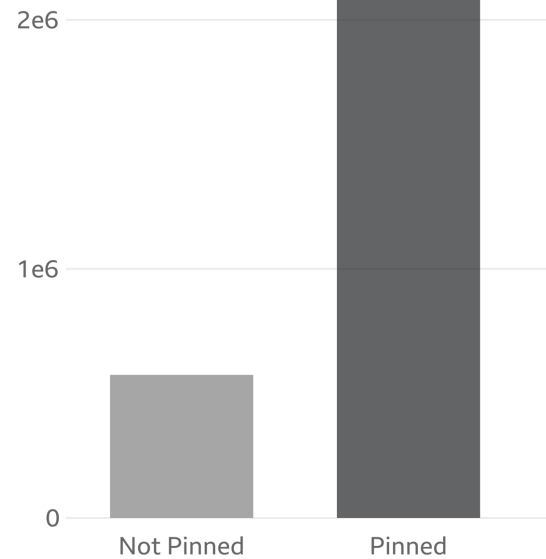
Grey areas are periods where Americast is pinned



% users in segment to click programme

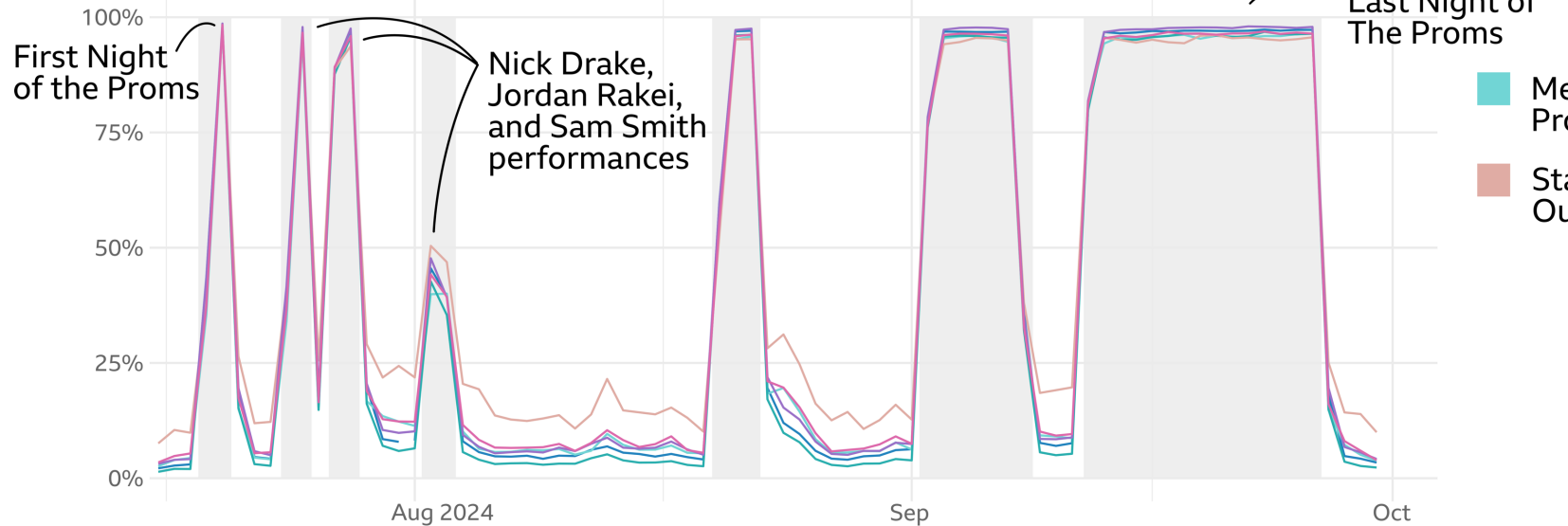


Avg Commonality Score



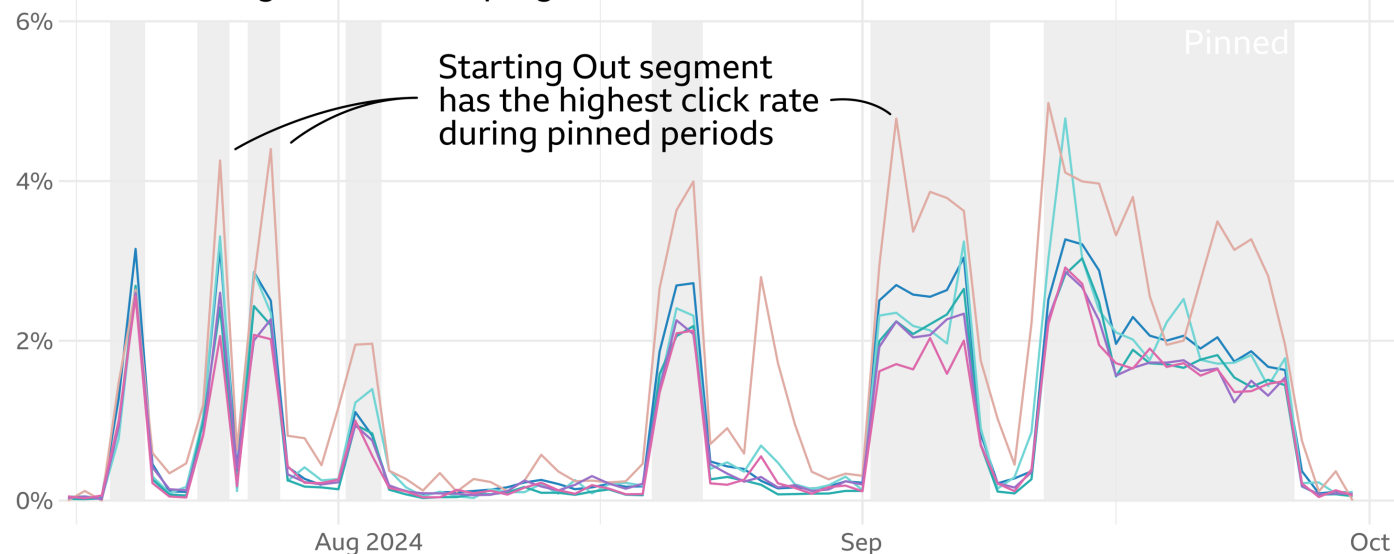
The Proms

% users in segment shown programme

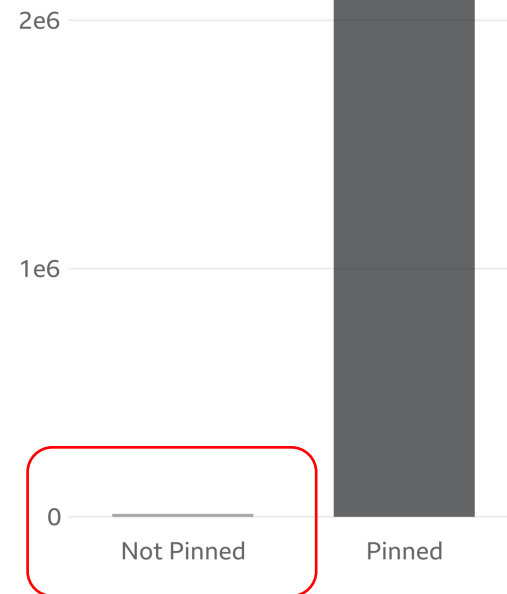


- Metropolitan Professionals
- Secure Suburbs
- Comfortable Retirees
- Starting Out
- Working Families
- Traditional Values

% users in segment to click programme



Avg Commonality Score



Rarely recommended without intervention!

2 further developments out of the BBC work – 2nd stage – BBC collaboration on commonality

- In assessing the utility of personalization and commonality, **further research is needed on their medium and long term impacts on audiences**: our study only assesses short-term impacts, neglecting important longer-term effects
- Pinning produces short term effects > research needed on **whether other techniques can be developed to achieve cumulative building of commonality?**
- Aim to develop commonality in **additional ways to target/cultivate niche, local and minority audiences with common content** (Fraser's 'counter-publics')
- ... And aim to extend this **fruitfully to news provision**, under editorial controls, with some news niche and some universal? And similar mix for other genres?
- But still limited by working with historical audience data
- ... And BBC's highly hierarchical: getting this work picked up by executives is difficult! (New CEO / DG ex-Google; £500m cuts just announced...)

Scaling up from the BBC collaboration – international initiative: Public Service Media in the Platform Era – Towards Public Digital Infrastructures / a Public Stack

This summit, hosted by the British Academy on Sept 3-4, addresses the acute challenges posed to public service media (PSM) worldwide by the global technology companies' dominance in the media ecosystem. This dominance in infrastructure, distribution platforms, algorithms and social media creates serious difficulties of **dependency**, exacerbated by lack of transparency and accountability. The global dominance of commercial platforms highlights the need to **ensure a robust ecosystem committed to public interest values, one that offers citizens genuine choices.**

The summit will **share evidence-based perspectives and case studies** showing how the challenges might be addressed in the UK and internationally.

Guiding principles to be debated by the conference are:

- 1) that it is only through international collaboration that sufficient resources can be built to support the development of **non-commercial, public-interest platforms and technologies at scale**
- 2) that such work is both feasible and underway, and needs to be shared
- 3) that the creation of a public service distribution platform and public service algorithms has the potential to bring significant economic benefits for the creative economies of the countries involved

We will debate these ideas with the industry, politicians, policy and academic communities concerned with PSM, and lay the groundwork for coordinated international action. We will showcase international speakers who will speak to what is possible in practice.

The summit is hosted by the UK's British Academy, and is organised by Georgina Born (UCL), Justin Lewis (Cardiff), Robin Mansell (LSE) Mark Andrejevic (Monash), the ERC MusAI programme, the Australian [ADM&S](#), the UKRI's [BBC-linked Responsible Innovation Centre](#), and the Public Media Alliance

Topics for discussion

- Challenges and limits of **interdisciplinarity** between CS & SSHA?
- **Translating normative principles into design** of AI systems (MRS, music database)?
.. And where do we find such principles?
- **Pluralising the political economy of AI beyond commercial tech**: what institutional and intellectual conditions make it possible to engage in alternative AI system design? Public, non-profit, alternative formations?
- Cultural (and music) **curation**: doing this differently?
- From shared **exposure** (common cultural experience) to **publics**: how could this be achieved? And how measured?
- **Beyond metrics**, what other interventions in MRS would also change listening culture?